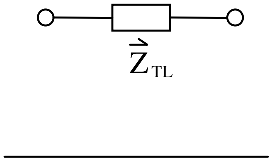
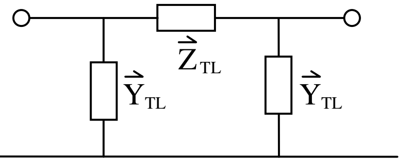
ENEE3522 – Transmission Line Modeling Formula Sheet

Transmission Line Modeling and Calculations

Series Impedance Model



Pi Model



$1\phi:\quad \vec{z} = \vec{z}_x + \vec{z}_y$

$3\phi:\quad \vec{z} = \vec{z}_A = \vec{z}_B = \vec{z}_C$

$\vec{Z}_{TL} = \vec{z}\,\ell \quad (\Omega)$

$\vec{Y}_{TL} = \frac{1}{2}\,\vec{y}\,\ell \quad (\mho)$

Distributed Series Impedance: $\vec{z} = R + j\omega L \quad (\Omega / \text{length unit})$ *Distributed Shunt Admittance*: $\vec{y} = G + j\omega C \quad (\mho / \text{length unit})$

Propagation Constant: $\vec{\gamma} = \sqrt{\vec{z}\cdot\vec{y}} \quad (/\text{length unit})$ *Characteristic Impedance*: $\vec{Z}_c = \sqrt{\vec{z}/\vec{y}} \quad (\Omega)$

Single Phase Transmission Line Impedance Parameter Calculations

$$R = \left(\frac{R_{sx}}{n_x} + \frac{R_{sy}}{n_y}\right) \qquad L = k_{L1}\cdot\ln\left(\frac{GMD_{xy}}{\sqrt{GMR_x\cdot GMR_y}}\right) \qquad C = k_{C1}\Big/\ln\left(\frac{GMD_{xy}}{\sqrt{mGMR_x\cdot mGMR_y}}\right)$$

$$GMD_{xy} = \left[\prod_{i=1}^{n_x}\left(\prod_{j=1}^{n_y} D_{xiyj}\right)\right]^{\frac{1}{n_x n_y}} \qquad k_{L1} = 6.437376\times10^{-4}\ H/mile \qquad k_{C1} = 4.47659\times10^{-8}\ F/mile$$

Three Phase Transmission Line Impedance Parameter Calculations

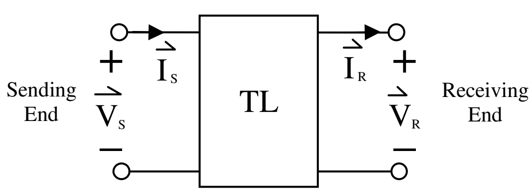
$$R = \left(\frac{R_s}{n}\right) \qquad L = k_{L3}\cdot\ln\left(\frac{GMD_{eq}}{GMR}\right) \qquad C = k_{C3}\Big/\ln\left(\frac{GMD_{eq}}{mGMR}\right)$$

$$GMD_{eq} = \sqrt[3]{D_{AB}D_{BC}D_{CA}} \qquad k_{L3} = 3.218688\times10^{-4}\ H/mile \qquad k_{C3} = 8.953181\times10^{-8}\ F/mile$$

Single Phase and Three Phase Transmission Line Conductor GMR Calculations

Single Cable (no bundling)	$GMR = GMR_s$	$mGMR = r_s$
Standard 2-Cable Bundle	$GMR = \sqrt{\mathrm{d}\cdot GMR_s}$	$mGMR = \sqrt{\mathrm{d}\cdot r_s}$
Standard 3-Cable Equilateral Bundle	$GMR = \sqrt[3]{\mathrm{d}^2\cdot GMR_s}$	$mGMR = \sqrt[3]{\mathrm{d}^2\cdot r_s}$
Standard 4-Cable Square Bundle	$GMR = \sqrt[4]{\sqrt{2}\cdot\mathrm{d}^3\cdot GMR_s}$	$mGMR = \sqrt[4]{\sqrt{2}\cdot\mathrm{d}^3\cdot r_s}$
For all Non-Standard Bundles	$GMR = \left[\prod_{i=1}^{n}\left(\prod_{j=1}^{n} D_{ij}\right)\right]^{\frac{1}{n^2}}$ $\quad mGMR = \left[\prod_{i=1}^{n}\left(\prod_{j=1}^{n} D^*_{ij}\right)\right]^{\frac{1}{n^2}}$ $D_{ij} = D^*_{ij} =$ *distance between subconductor i and subconductor j* $D_{ii} = GMR_S,\ D^*_{ii} = r_s$ *for both stranded and solid cables* $GMR_S =$ *table* GMR_S *for stranded cables, or* $GMR_S = e^{-0.25} r_s$ *if solid cables*	

Transmission Line 2-Port Network Modeling and Equations



Forward Equations

$$\begin{cases}\vec{V}_S = \vec{A}\vec{V}_R + \vec{B}\vec{I}_R\\ \vec{I}_S = \vec{C}\vec{V}_R + \vec{D}\vec{I}_R\end{cases}$$

Reverse Equations

$$\begin{cases}\vec{V}_R = \vec{A}\vec{V}_S - \vec{B}\vec{I}_S\\ \vec{I}_R = -\vec{C}\vec{V}_S + \vec{D}\vec{I}_S\end{cases}$$

Short Length (Series Impedance) Model	$\vec{A} = \vec{D} = 1, \quad \vec{B} = \vec{Z}_{TL}, \quad \vec{C} = 0$
Medium Length (Pi) Model	$\vec{A} = \vec{D} = 1 + \vec{Z}_{TL}\vec{Y}_{TL}, \quad \vec{B} = \vec{Z}_{TL}, \quad \vec{C} = \vec{Y}_{TL}\left(2 + \vec{Z}_{TL}\vec{Y}_{TL}\right)$
Long Length Model	$\vec{A} = \vec{D} = \cosh\left(\vec{\gamma}\ell\right), \quad \vec{B} = \vec{Z}_c\sinh\left(\vec{\gamma}\ell\right), \quad \vec{C} = \vec{Z}_c^{-1}\sinh\left(\vec{\gamma}\ell\right)$ $\begin{cases}\vec{V}(x) = \left[\cosh\left(\vec{\gamma}x\right)\right]\vec{V}_R + \left[\vec{Z}_c\sinh\left(\vec{\gamma}x\right)\right]\vec{I}_R\\ \vec{I}(x) = \left[\vec{Z}_c^{-1}\sinh\left(\vec{\gamma}x\right)\right]\vec{V}_R + \left[\cosh\left(\vec{\gamma}x\right)\right]\vec{I}_R\end{cases}$